

G11 PHYSICS

Assignment 03

Due: 13 April 2026

1. Find the derivative of the following functions.

a. $f(t) = t + 1$

c. $f(t) = t^5 + 6t$

b. $f(x) = bx + x^{1/2}$

d. $f(x) = 4 + 8x - 5x^2$

2. Compute your average velocity if you walk 15 m at a speed of 3 m/s and then run 35 m at a speed of 9 m/s along a straight track.

3. A cliff diver plunges from a height of 80 m above the water surface. The distance the diver falls in t seconds is given by $s(t) = 4.9t^2$ (in meters).

(i) After how many seconds does the diver hit the water?

(ii) With what velocity does the diver hit the water?

4. Find the integral of the following functions.

a. $f(t) = 1$

c. $f(t) = 5t^4 + 6$

b. $f(x) = b + \frac{1}{2}x^{-1/2}$

d. $f(x) = 8 - 10x$